Investigation: RECRUSUL S.A. (CVM code 12572), fiscal year 2015, as it was on 2024-12-31

Statistical signal based on public financial statements. Not an accusation, not investment advice.

Company

Sector: Máquinas, Equipamentos, Veículos e Peças (origin: FCA); group: non-financial.
Before 2018, the company was called RECRUSUL SA.
Trading codes (2023 FCA, version 1, filed on 2023-05-08): RCSL3 (Ações Ordinárias, Bolsa, since 1985-12-20); RCSL4 (Ações Preferenciais, Bolsa, since 1985-12-20).

Document used

2015 DFP, version 1, filed on 2016-08-15; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2016-08-15).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=59130

Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: −6.164 (index)
- Alert by the study 1 rule: no
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: 8 (n = 18)
- Percentile in group and year: 3 (n = 376)
- The company's own history: median −2.749; Tukey range (1.5 × IQR): from −10.148 to 2.734; value inside the range (index; 5 fiscal years)
- Confidence in the data: medium

DSRI, days' sales in receivables index
- Value: 0.000 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 2 (n = 21)
- Percentile in group and year: 1 (n = 460)
- The company's own history: median 1.019; Tukey range (1.5 × IQR): from 0.016 to 2.545; value below the range (index; 5 fiscal years)
- Confidence in the data: medium

GMI, gross margin index
- Value: 0.591 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 12 (n = 21)
- Percentile in group and year: 10 (n = 481)
- The company's own history: median 1.024; Tukey range (1.5 × IQR): from 0.670 to 1.547; value below the range (index; 5 fiscal years)
- Confidence in the data: medium

AQI, asset quality index
- Value: 1.055 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 55 (n = 21)
- Percentile in group and year: 67 (n = 464)
- The company's own history: median 0.959; Tukey range (1.5 × IQR): from 0.364 to 1.661; value inside the range (index; 5 fiscal years)
- Confidence in the data: medium

SGI, sales growth index
Value: 0.288 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 7 (n = 21)
Percentile in group and year: 2 (n = 483)
The company's own history: median 1.277; Tukey range (1.5 × IQR): from −1.561 to 3.290; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

DEPI, depreciation index
Value: 1.327 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 92 (n = 18)
Percentile in group and year: 94 (n = 393)
The company's own history: median 0.971; Tukey range (1.5 × IQR): from 0.403 to 1.373; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

SGAI, SG&A expenses index
Value: 0.726 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 12 (n = 21)
Percentile in group and year: 9 (n = 470)
The company's own history: median 0.904; Tukey range (1.5 × IQR): from −2.288 to 6.120; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

TATA, total accruals to total assets
Value: −0.417 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 12 (n = 21)
Percentile in group and year: 4 (n = 489)
The company's own history: median −0.044; Tukey range (1.5 × IQR): from −0.930 to 0.525; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

LVGI, leverage index
Value: 1.203 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 88 (n = 21)
Percentile in group and year: 82 (n = 484)
The company's own history: median 1.160; Tukey range (1.5 × IQR): from −0.527 to 3.223; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

DPO jump
Value: 4,041.1 days
Alert by the study 1 rule: yes
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 42.4 days)
Percentile in sector and year: 98 (n = 21)
Percentile in group and year: 99 (n = 449)
The company's own history: median −31.7 days; Tukey range (1.5 × IQR): from −489.6 days to 693.8 days; value above the range (5 fiscal years)
Confidence in the data: medium

Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: 64.01 p.p.
Alert by the study 1 rule: yes
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 88 (n = 21)
Percentile in group and year: 92 (n = 438)
The company's own history: median 125.18 p.p.; Tukey range (1.5 × IQR): from 90.36 p.p. to 147.00 p.p.; value below the range (5 fiscal years)
Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −0.406 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 12 (n = 21)
Percentile in group and year: 4 (n = 488)
The company's own history: median −0.045; Tukey range (1.5 × IQR): from −0.848 to 0.475; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

Financial risk (context only)

Altman Z" (financial risk, context only)
Value: −32.948 (index)
Components: X1 = −2.169; X2 = −5.191; X3 = −0.151; X4 = −0.747 (indices)
Percentile in sector and year: 7 (n = 22)
The company's own history: median −7.471; Tukey range (1.5 × IQR): from −14.109 to −2.420; value below the range (index; 5 fiscal years)
- No cut-off zone: the available source does not define zones for Z"
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 2/9 (8 computable)
Signals:
  - Return on assets (ROA): 0
  - Operating cash flow on assets (CFO): 0
  - Change in ROA: 0
  - Cash above earnings (ACCRUAL): 1
  - Change in leverage: 0
  - Change in current ratio: 0
  - No equity issuance (EQ_OFFER): —
  - Change in gross margin: 1
  - Change in asset turnover: 0
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

Changes

Restatement through the following year's comparative: no difference above the tolerance, or the next year's DFP did not exist yet on the date.

What was known on the date

- DFP 2010, v1: 2011-03-29
- DFP 2011, v1: 2012-03-30
- DFP 2012, v1: 2013-03-27
- DFP 2012, v2: 2013-04-04
- DFP 2013, v1: 2014-03-31
- FCA 2015, v1: 2015-03-03
- FCA 2015, v2: 2015-05-15
- DFP 2014, v1: 2015-06-30
- FCA 2015, v3: 2015-09-03
- DFP 2015, v1: 2016-08-15
- DFP 2016, v1: 2017-03-31

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

Data quality

53 applicable checks; 0 failed.

Auditor's report and auditor (factual section; no alert)

Auditor's report: Sem Ressalva (unqualified).
no opinion excerpt was found in the text
Auditor in the 2021 FCA: Taticca Auditores Independentes SS.
Auditor in the 2022 FCA: Taticca Auditores Independentes SS.
Auditor in the 2023 FCA: Taticca Auditores Independentes SS.
Auditor changes (one line per change, canonical name by the firm's CVM code):

- 2012: DRS Auditores -> Michelon & Cia Auditores e Consultores SS. The outgoing firm audited 5 consecutive fiscal years (2007 to 2011), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).
- 2015: Michelon & Cia Auditores e Consultores SS -> Baker Tilly Brasil RS Auditores Independentes Sociedade Simples. The outgoing firm audited 3 consecutive fiscal years (2012 to 2014), by the start and end dates in the FCA.
- 2020: Baker Tilly Brasil RS Auditores Independentes Sociedade Simples -> Taticca Auditores Independentes SS. The outgoing firm audited 5 consecutive fiscal years (2015 to 2019), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

Document excerpts (copied; no alert)

Excerpts copied from the documents, with the page, and the Reference Form table. It is not a signal family: it generates no alert, score or ranking. Absence of mention does not prove absence.
How the reader was tested: 20 held-out cases, a single measurement with the frozen reader (c99c953): precision 0.962, coverage 0.962, literal check on 132 of 132 excerpts, no natural-

person name (docs/leitor_medida.json).

Explanatory notes · Related parties
Note: NOTA 07 - TRANSAÇÕES ENTRE PARTES RELACIONADAS (pages 7 to 8)
- [subsidiaries] “Créditos em Controladas Ativo Não Circulante - - - - - 7.901” (p. 7, original excerpt in Portuguese)
- Excerpt omitted; see page 7 of the document
Fields of item 18 of CPC 05 (R1)
- [balances] “Créditos em Controladas Ativo Não Circulante - - - - - 7.901” (p. 7, original excerpt in Portuguese)
- [balances; allowance for doubtful debts] “(-) Provisão para Perdas Ativo Não Circulante - - - - - (7.901)” (p. 7, original excerpt in Portuguese)
- Excerpt omitted; see page 7 of the document
- [balances; allowance for doubtful debts] “(-) Provisão para Perdas Passivo Não Circulante - - - - - (7.901)” (p. 7, original excerpt in Portuguese)

Explanatory notes · Reverse factoring and forfaiting
The reader found no mention of reverse factoring (risco sacado), forfaiting or supplier agreements in the notes of this DFP. Absence of mention does not prove absence of the operation.

Auditor's report
Type of opinion: unqualified
- Excerpt omitted; see the auditor's report in the document
- [other matters] “Outros assuntos Demonstrações do valor adicionado Examinamos, também, as demonstrações individuais e consolidadas do valor adicionado (DVA), referentes ao exercício findo em 31 de dezembro de 2015, elaboradas sob a responsabilidade da administração da Companhia, cuja apresentação é requerida pela legislação societária brasileira para companhias abertas, e como informação suplementar pelas IFRS que não requerem a apresentação da DVA. Essas demonstrações foram submetidas aos mesmos procedimentos de auditoria descritos anteriormente e, em nossa opinião as demonstrações contábeis individuais e consolidadas estão adequadamente apresentadas, em todos os seus aspectos relevantes, em relação às demonstrações contábeis tomadas em conjunto.” (original excerpt in Portuguese)

Reference Form
Form of 2016, version 6, filed on 2017-08-22.
It is the only version available in the open data; earlier versions are not there.
- Related party (name omitted) · Controlada · Pagamento de despesas · 384374.43 · 384.374,43
- Related party (name omitted) · Controlada · Pagamento de despesas · 1273860.51 · 1.273.860,51
- Related party (name omitted) · Controlada · Empréstimo de controlada · 23390.42 · 23.390,42

Limits of this section:
- In real documents, about 22% of the excerpts come out omitted by the name detector, including names of companies outside the registry.
- In tables, header lines (years, column titles) may come out as excerpts; the held-out cases had no tables with multi-line headers.

How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial

companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 38 DFP lines, each with document, version, filing date, account and location.

Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a F1); leitor-f1-1 (commit c99c953). Generated at 2026-11-12T03:59:02+00:00.

Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.
Name in today's registry: RECRUSUL SA.